



Oxidative stress response and gene expression with atrazine exposure in adult female zebrafish (*Danio rerio*)

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ABSTRACT

Atrazine (ATZ) is the most common pesticide of freshwater ecosystems in the world. The effects of ATZ exposure on the induction of oxidative stress and the alteration of gene expression were studied in liver and ovary samples from female zebrafish (*Danio rerio*). Antioxidant enzyme activities (SOD and CAT), in addition to the GSH and MDA content, in the liver altered significantly; the mRNA levels for the genes encoding these antioxidant proteins, such as Cu/Zn-Sod, Mn-Sod, Cat, and Gpx, were up-regulated significantly in the liver when zebrafish were exposed to various concentrations of ATZ for 14 d. However, the mRNA induction patterns were not in accordance with those of antioxidant enzyme changes in the liver and ovary. Moreover, the transcriptional expression of mitochondrial inner membrane genes related to ROS production, such as Ucp-2 and Bcl-2, were altered significantly in high ATZ treatment groups. However, the ATZ treatment did not induce any apparent changes in the transcriptional expression of Cox1, Ndi, or ATP6, which are related to the mitochondrial respiratory chain and ATP synthesis. To our knowledge, this is the first report to study the transcriptional effects of ATZ stress on zebrafish. The information presented in this study will be helpful in fully understanding the mechanism of oxidative stress induced by ATZ in fish.

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1. Introduction

Many freshwater ecosystems are contaminated with industrial, domestic and agricultural chemicals such as herbicides, which are ubiquitous and can spread regionally and globally (Flynn and Spellman, 2009). In the last 40 years, atrazine (2-chloro-4-ethylamino-6-isopropylamino-s-triazine, ATZ) has been one of the most extensively used herbicides. It is primarily applied to increase the yield of food crops such as soy, maize and sugarcane, and is the most commonly used pesticide in the United States. According to USDA NASS (2008), in 2003 about 24.3 million kg of ATZ was used on corn crops in the United States. In China, where ATZ is used as a broad-spectrum herbicide, which is applied to major field crops such as corn, wheat and soybeans, it has been reported that in 2008 the amounts of ATZ applied to fields was at least 5000 t (Song et al., 2009). Runoff and erosion are the major routes of ATZ entry into surface waters, while leaching and lateral movement through the soil or tile drains are secondary routes of entry (Solomon et al., 2008). As a result, ATZ is commonly monitored in groundwater and surface water (Gfrerer et al., 2002); and misuses of ATZ and other pesticides can have effects on non-target organisms.

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Oxidative stress has become an important subject in aquatic toxicology (Livingstone, 2001), and ATZ may be directly involved in this process. ATZ may be responsible for induction of oxidative stress, by interfering with different endpoints and ROS production in some fish species. Reactive oxygen species (ROS), such as superoxide anion radicals, hydrogen peroxide, and hydroxyl radicals, are continuously formed in oxygen-consuming organisms. Exposure to xenobiotics or toxic chemical pollutants may produce an imbalance between these endogenous and exogenous ROS, and can subsequently induce a decrease in antioxidant defenses or cause oxidative damage outright in organisms (Valavanidis et al., 2006). Fish, like many other vertebrates, try to reduce the damage from oxidative stress using an antioxidant defense system; the first line of defense consists of antioxidant molecules, such as glutathione (GSH), vitamin C and E, carotenoids (Alvarez et al., 2005). Another defensive mechanism is comprised of antioxidant enzymes, including the following radical-scavenging enzymes: superoxide dismutase (SOD), catalase (CAT), glutathione peroxidase (GPx), etc. (Valavanidis et al., 2006). Recently, measurements of antioxidative enzyme activities in fish have been used to assess the oxidative damage caused by chemicals in aquatic ecosystems (Zhang et al., 2009). Correspondingly, a number of studies have shown that ATZ can promote oxidative stress, by increasing the concentration of ROS and products of oxidative damage (for example, lipid

peroxides), and by influencing antioxidant enzyme activities in some organisms. Elia et al. (2002) demonstrated that various concentrations of ATZ (3, 6 and 9 mg L⁻¹) could increase antioxidant enzyme of SOD and the GSH content in the liver of bluegill fish after 96 h of exposure. These findings suggest that sub-acute ATZ concentrations can induce oxidative stress in bluegill fish. More recently, Song et al. (2009) reported that the SOD activity was stimulated under 2.5 mg kg⁻¹ ATZ treatment and inhibited under 5, 10 mg kg⁻¹ treatments in root of *Vicia faba*, and even induced ROS production and DNA damage.

Although it is likely that ATZ may influence the fish antioxidant system, the molecular mechanism has only received limited attention. Specifically, little is known about the effects of oxidative stress on the antioxidant activity, mitochondrial ROS production and respiratory chain related gene transcription induced by ATZ in any fish species. Thus, to better understand the oxidative stress process induced by ATZ, the present study examines concentration-dependent oxidative stress in the female adult zebrafish liver and ovary by determining the malondialdehyde (MDA) and GSH contents and the activities of SOD and CAT. In addition, the mRNA expression levels of genes encoding antioxidant proteins, such as Cu/Zn-superoxide dismutase (Cu/Zn-Sod), manganese superoxide dismutase (Mn-Sod), Cat, Gpx, were examined. Moreover, the transcriptional expression of mitochondrial inner membrane genes related to ROS production, the respiratory chain, and ATP synthesis, such as uncoupling protein 2 (Ucp-2), B-cell lymphoma/leukemia-2 gene (Bcl-2), NADH dehydrogenase subunit (Ndi), cytochrome c oxidase subunit I (CoxI), and ATP synthase F0 subunit 6 (ATPo6), in the liver were also investigated. This information is intended to provide new insights into the toxicological mechanism of ATZ.

2. Materials and methods

2.1. Experimental fish

Healthy five-month old adult female fish were selected and kept in aquaria. Prior to exposure experiments, the fish were acclimatized in a tank filled with water at ambient temperature (25 ± 1 °C) for one week, with a photoperiod consisting of 14-h light/10-h dark segments for each day. The fish were fed twice a day with brine shrimp. Female (weight 0.52 ± 0.08 g) fish were randomly selected for exposure experiments. There were no statistically significant differences in body weight or length at the beginning of exposure (data not shown).

2.2. Fish exposure and sample collection

ATZ (CAS No.: 1912-24-9) was purchased from Sigma–Aldrich (USA) and used as received. A stock solution of ATZ was prepared by dissolving the solid in acetone at 1 g L⁻¹. To obtain a better assessment of oxidative stress, female adult zebrafish were exposed to ATZ at various concentrations (1 10 100, and 1000 µg L⁻¹) in water containing 0.2% acetone. The water was dechlorinated and prefiltered through activated carbon prior to use (pH 6.7–7.6, hardness 40–60 mg CaCO₃ L⁻¹). Ten female adult fish were reared in 3 L of each solution in an aquarium, and four separated glass aquaria were used for each treatment. Control female fish were raised in rearing water containing 0.2% acetone in the absence of ATZ. Zebrafish were exposed to the test solutions under static conditions for 14 d. Rearing water and exposure solutions were completely changed (100%) every day.

Before dissection, the fish were anesthetized on ice. The livers and ovaries excised from the fish in each exposure vessel were randomly divided into two samples: at least four fish were collected as

one sample, resulting in four pooled samples for enzyme extraction and biochemical analysis; and four other samples were pooled for RNA extraction and mRNA transcription analysis. These samples were kept on dry ice while being prepared and then stored at –80 °C until they were analyzed. No fish died during the course of the exposure.

2.3. Enzyme extraction and biochemical assays

Liver and ovary samples were defrosted and homogenized on ice with 10 volumes of cold buffer consisting of 250 mmol L⁻¹ sucrose, 5 mmol L⁻¹ Tris–HCl, and 0.1 mmol L⁻¹ Edetic acid-2Na (pH 7.5). The homogenate was centrifuged at 4000g at 4 °C for 15 min to obtain the supernatant for the enzyme activity assays. The supernatant was used to assay for lipid peroxidation and GSH content, in addition to SOD and CAT activities, using commercial kits (Nanjing Jiancheng Bioengineering Institute, Nanjing, China).

The methods used to determine the content of MDA and GSH, and the activities of SOD and CAT, in zebrafish tissue were performed using the instructions from the kits and the established method described by Liu et al. (2008). Briefly, lipid peroxidation products (measured as MDA) were quantified using the thiobarbituric acid (TBARS) method, and the MDA concentration was expressed as nmol mg⁻¹ protein. The GSH content was expressed as mg g⁻¹ protein. One unit of SOD activity was defined as the amount of enzyme required to inhibit the oxidation reaction by 50% and was expressed as U mg⁻¹ protein. One unit of CAT activity was defined as the amount of enzyme required to consume 1 µmol H₂O₂ in 1 s and was expressed as U mg⁻¹ protein. Protein concentration was determined using bicinchoninic acid (BCA) as a detection reagent for Cu⁺ following the reduction of Cu²⁺ by protein in an alkaline environment (BCA protein kit, Sangon company, China). Measurements were made on a microplate reader, using micro well plate protocol A590, according to the manufacturer's instructions.

2.4. Gene expression analysis

Total RNA was isolated from liver using TRIzol reagent (Takara Biochemicals, Dalian, China) according to the manufacturer's protocol. The ratio of absorbance at 260 nm to that at 280 nm, as well as the banding pattern on a 1% agarose formaldehyde gel, was used to verify the quality of the RNA in each sample. Subsequently, the RNA was denatured at 65 °C for 15 min. cDNA was synthesized using the M-MLV reverse transcriptase kit (Toyobo, Tokyo, Japan). A 1 µL portion of the RT products was used directly for real-time polymerase chain reaction (PCR). Real-time quantitative PCR was performed on an ABI-7300 instrument (Advanced Biosystems, Foster, California, USA). Oligonucleotide primers were used to detect the gene expression of β-actin, Cu/Zn-Sod, Mn-Sod, Cat, Gpx, Bcl-2, Ucp-2, Ndi, CoxI, and ATPo6 using the SYBR green system (Toyobo, Tokyo, Japan); detailed information is shown in Table 1. As a house-keeping gene, β-actin transcript was used to standardize the results by eliminating variations in mRNA and cDNA quantity and quality, and each mRNA level was expressed as its ratio to β-actin mRNA. PCRs were performed three times, in separate runs. The following PCR protocol was used with the ABI-7300: denaturation for 1 min at 95 °C, followed by 40 cycles of 15 s at 95 °C, and 1 min at 60 °C. The relative quantification of gene expression among the treatment groups was analyzed by the 2^{-ΔΔCt} method (Livak and Schmittgen, 2001).

2.5. Data analysis

The experimental data were checked for normality and homogeneity of variance using the Kolmogorov–Smirnov one-sample

Table 1

Sequences of primer pairs used in the real-time quantitative PCR reactions.

Target gene	Primer sequences	Product length (bp)	Accession number
Mn-Sod	F: 5'-CCGGACTATGTTAAGGCCATCT-3' R: 5'-ACACTCGGTGCTCTCTTTCTCT-3'	123	AY195857
Cu/Zn-Sod	F: 5'-GTCGTCTGGCTTGGAGTG-3' R: 5'-TGTCAGCGGGCTAGTGCTT-3'	113	Y12236
Cat	F: 5'-AGGGCAACTGGGATCTTACA-3' R: 5'-TTTATGGGACCAGACCTTGG-3'	499	AF170069
Gpx	F: 5'-AGATGTCATTC2TCGACACG-3' R: 5'-AAGGAGAAGCTTCCTCAGCC-3'	94	AW232474
Ucp-2	F: 5'-TGGCTCAACCCACTGATGTA-3' R: 5'-CAATGGTCCGATATGCGTC-3'	102	NM_131176
Bcl-2	F: 5'-AGGAAAATGGAGGTGGGATG-3' R: 5'-TGTTAGGTATGAAAACGGGTGGA-3'	83	AY695820
Ndi	F: TACAGAGGGGGAATCAGAAC R: TTGGTCGTATCGGAATCGT	253	AC024175
CoxI	F: GGATTTGGAACTGACTTGTC R: AAGAAGAAATGAGGGTGAAG	105	AC024175
ATPo6	F: TTATCCTCGTTGCCATCTTC R: AGTTGGTTTGTAATCGTCC	118	AC024175
β -Actin	F: 5'-ATGGATGAGGAAATCGCTGCC-3' R: 5'-CTCCCTGATGTCTGGGTCGTC-3'	106	AF057040

test and Levene's test, respectively. When necessary, the data were transformed for purposes of normalization and homogeneity of variance. Intergroup differences were assessed using analysis of variance (ANOVA) followed by Dunnett's test, using the StatView 5.0 program (SAS Institute Inc., Cary, NC, USA). Values were considered statistically significant when p was less than 0.05 or 0.01. The figures were composed using Origin 8.0 (OriginLab, Northampton, MA, USA).

3. Results

3.1. Effect of ATZ on antioxidant enzymes and GSH content

Fig. 1 shows that the activities of the antioxidant enzymes SOD and CAT increased in a dose-dependent manner in the liver of female zebrafish after a 14 d ATZ exposure. SOD activity increased significantly at higher treatment concentrations; it was about 1.51-fold and 1.62-fold higher than those of the control for 100 and 1000 $\mu\text{g L}^{-1}$ ATZ, respectively (Fig. 1A). With respect to CAT activity, only treatment with the highest concentration of ATZ induced a significant increase (Fig. 1B). The induction of SOD and CAT activities was also observed in the ovary of zebrafish, but the processes were different from that observed in the liver. Maximum induction of SOD and CAT activities was only apparent with a treatment concentration of 10 $\mu\text{g L}^{-1}$ ATZ, when SOD and CAT activities increased by 58.1% and 61.7%, respectively, compared to the control group (Fig. 1D and E). Correspondingly, a dose-dependent decrease in GSH was observed, particularly in the liver of zebrafish. Exposure to higher concentration (100 and 1000 $\mu\text{g L}^{-1}$) of ATZ resulted in a significant decrease in both the liver and ovary GSH levels, and the highest decrease occurred at the treatment of 1000 $\mu\text{g L}^{-1}$ group compared to the control fish (decreased 39% and 20.3% respectively) (Fig. 1C and F).

3.2. Effect of ATZ on lipid peroxidation

The results in Fig. 2 illustrate how MDA levels increase in a dose-dependent manner in the liver of female adult zebrafish. Significant increases were observed in the 100 and 1000 $\mu\text{g L}^{-1}$ ATZ treatments, where the MDA levels were about 1.8-fold and 2.1-fold higher than that of the control fish (Fig. 2A). However, exposure of adult female zebrafish for 14 d to each of the ATZ concentrations

tested in this study did not induce a significant increase in the level of MDA in ovarian tissues (Fig. 2B).

3.3. Effects of ATZ on mRNA expression

The effects of ATZ exposure on the mRNA levels of various genes (Cu/Zn-Sod, Mn-Sod, Cat and Gpx) encoding antioxidant proteins were determined by RT-qPCR (Fig. 3). The mRNA level of Mn-Sod increased by about 2-fold and 3.4-fold, in the liver of zebrafish, in the 10 and 100 $\mu\text{g L}^{-1}$ ATZ treatment groups, respectively (Fig. 3A). Similarly, Cu/Zn-Sod mRNA levels in the liver increased significantly in the 10 and 100 $\mu\text{g L}^{-1}$ ATZ treatment groups, with increases of about 3.67-fold and 3.6-fold, respectively, compared with the control group (Fig. 3B). However, both Cu/Zn- and Mn-Sod mRNA levels were not significantly up-regulated at the 1000 $\mu\text{g L}^{-1}$ ATZ treatment concentration. The kinetics of Cat and Gpx expression were similar, with significant levels of induction appearing in the 100 $\mu\text{g L}^{-1}$ ATZ treatment group (Fig. 3C and D). In addition, compared to the liver, no significant induction in any of the four selected genes was observed in the ovary (Fig. 3E–H), suggesting that the ovary is less sensitive to the oxidative stress induced by ATZ in zebrafish. More importantly, liver was considered as the primary organ to be responsible for biotransformation and detoxification of xenobiotics in fish (Hoffmann et al., 2008). For these reasons, we decided to detect the mRNA expression related to ROS production, the respiratory chain, and ATP synthesis in liver.

Fig. 4 shows the expression of mitochondrial inner membrane genes related to the regulation of ROS production, the respiratory chain, and ATP synthesis. Ucp-2 and Bcl-2, which are located in the inner membrane of mitochondria, play important roles in the regulation of ROS production and antioxidant. Bcl-2 mRNA expression was decreased in response to ATZ exposures at concentrations higher than 10 $\mu\text{g L}^{-1}$; at ATZ concentrations of 10 100 and 1000 $\mu\text{g L}^{-1}$, Bcl-2 mRNA levels were about 3.5-fold, 2.2-fold and 4.2-fold lower, respectively, compared to that of control samples (Fig. 4D). On the other hand, the expression of Ucp-2 was significantly up-regulated by 2.2-fold only in the 1000 $\mu\text{g L}^{-1}$ treatment group, compared to control (Fig. 4E). Moreover, the expression of Ndi, CoxI, and ATPo6, which are involved in the mitochondrial respiratory chain and ATP synthesis, exhibited no significant differences between the control and any of the ATZ-treated groups ($p > 0.05$) (Fig. 4A–C).

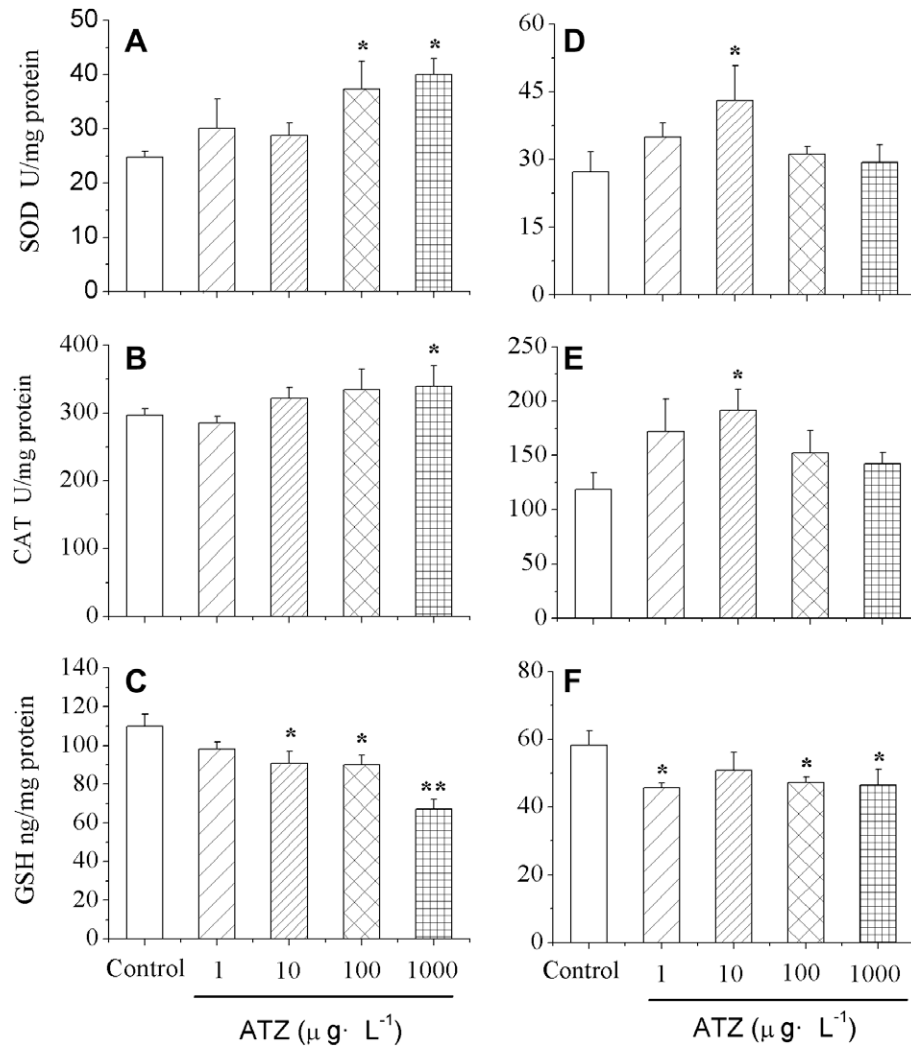


Fig. 1. Activities of SOD (A, D) and CAT (B, E), and GSH (C, F) in the liver (A, B and C) and ovary (D, E and F) of female zebrafish exposed to various concentrations of atrazine for 14 d. Values are presented as a mean $\pm$ SEM ($n = 4$). The asterisk represents a statistically significant difference when compared with the female controls; * at $p < 0.05$ and ** at $p < 0.01$ levels.

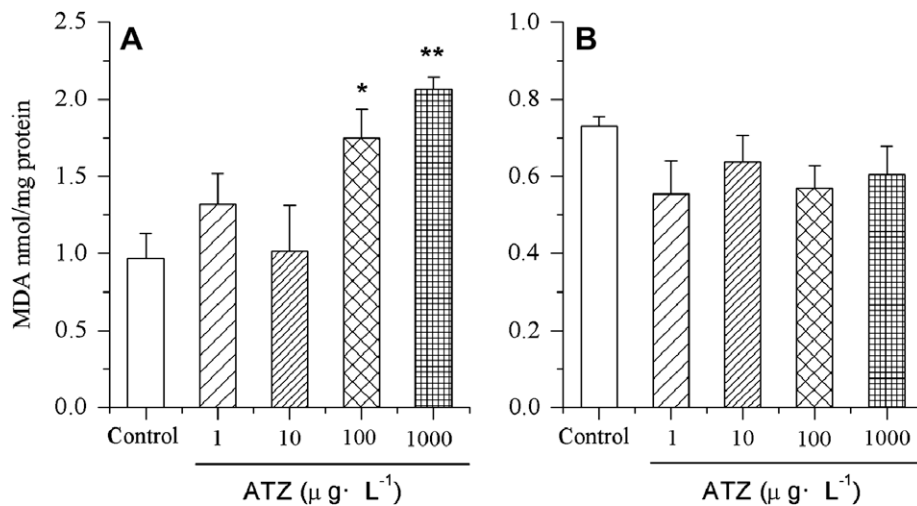


Fig. 2. MDA contents in the liver (A) and ovary (B) of female zebrafish exposed to various concentrations of atrazine for 14 d. Values are presented as a mean $\pm$ SEM ($n = 4$). The asterisk represents a statistically significant difference when compared with the female controls; * at $p < 0.05$ and ** at $p < 0.01$ levels.

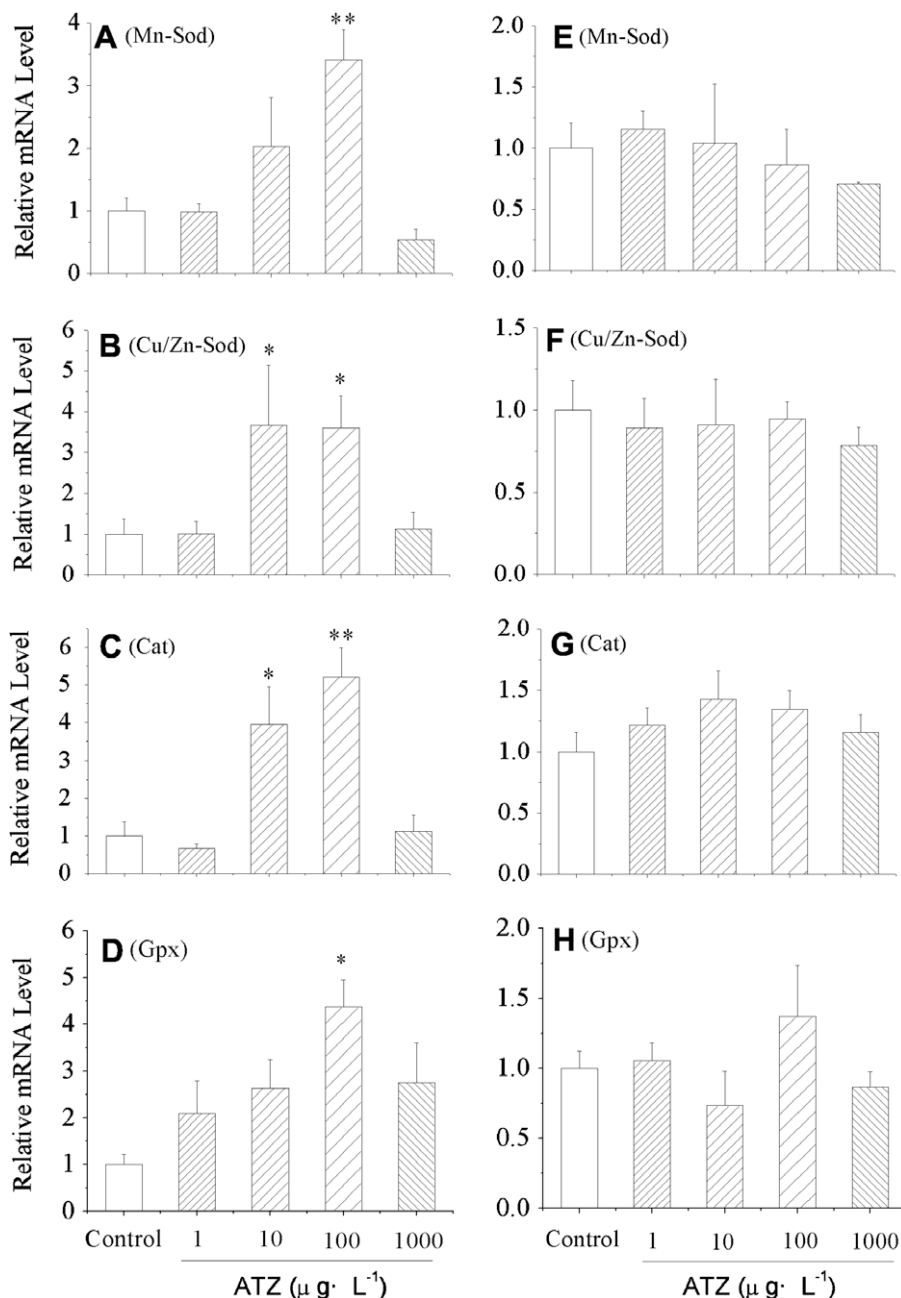


Fig. 3. Expression of Mn-Sod, Cu/Zn-Sod, Cat and Gpx in the liver (A, B, C, D) and ovary (E, F, G, H) of zebrafish exposed to various concentrations of ATZ for 14 d. Values were normalized against β -actin (used as a house-keeping gene), and represent the mean mRNA expression value $\pm$ SEM ($n = 4$) relative to those of the controls. The asterisk represents a statistically significant difference when compared with the female controls; *at $p < 0.05$ and **at $p < 0.01$ levels.

4. Discussion

The increase in lipid peroxidation following ATZ exposure may be attributed to the induction of ROS, which enhance the oxidation of polyunsaturated fatty acids and lead to lipid peroxidation (Valavanidis et al., 2006; Liu et al., 2008). In high ATZ concentration treatment groups, the MDA content, which is a secondary lipid peroxidation product (Draper et al., 1993), increased by more than two times compared to the control fish, suggesting that a large amount of lipid peroxidation has occurred in the treated fish.

Many organisms have unique systems for protecting themselves against the damaging effects of activated ROS. GSH is an important antioxidant and the amount of GSH present could reflect the antioxidant potential of an organelle. Likewise, the antioxidant

enzyme GPx catalyzes the reduction of hydrogen peroxide, but at the expense of GSH (Liu et al., 2008). In the present study, ATZ exposure induced a noticeable concentration-dependent decrease in GSH content in both the liver and ovary. Furthermore, SOD and CAT activities increased, especially in the liver, after ATZ exposure. Like other organisms, fish can combat the elevated levels of ROS in their systems with protective ROS-scavenging enzymes such as SOD and CAT, which convert superoxide anions (O_2^-) into H_2O_2 and then into H_2O and O_2 . Thus, it is possible that an increase in the activities of these enzymes contributes to the elimination of ROS from the cell induced by ATZ exposure. Similar results have also been reported in other species. For example, Qian et al. (2008) reported that the various concentrations (from 50 to 400 $\mu\text{g L}^{-1}$) of ATZ exposure induced a significant dose-dependent

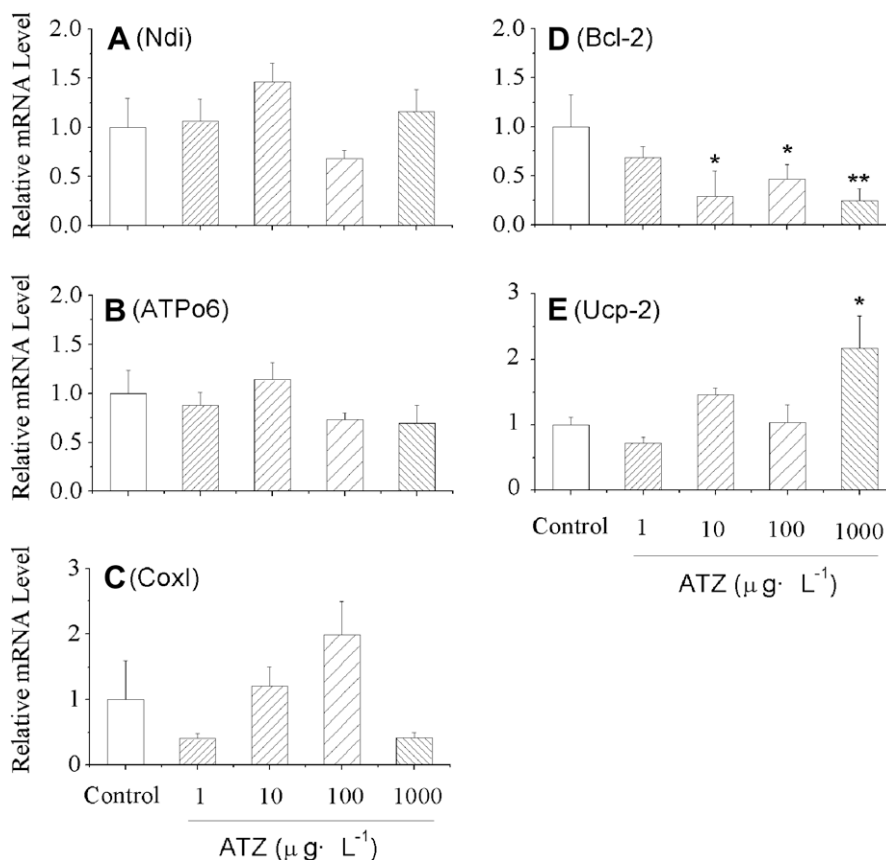


Fig. 4. Expression of Ndi, ATP6, CoxI, Bcl-2 and Ucp-2 in the liver (A, B, C, D, E) of zebrafish exposed to various concentrations of ATZ for 14 d. Values were normalized against β -actin (used as a house-keeping gene), and represent the mean mRNA expression value $\pm$ SEM ($n = 4$) relative to those of the controls. The asterisk represents a statistically significant difference when compared with the female controls; *at $p < 0.05$ and **at $p < 0.01$ levels.

increase in SOD and CAT activities in *Chlorella* sp. Recently, Salaberria et al. (2009) suggested that the activity of CAT, a biomarker of oxidative stress, was up-regulated significantly in rainbow trout after a single injection of ATZ (200 µg kg⁻¹). More importantly, in the present study, although the parameters of ROS-related enzymes were altered significantly, no metabolic or cellular dysfunction was observed correspondingly. In fact, cells are capable to deal with increases in ROS up to a certain point without any deleterious effect on cellular function or viability. Thus, the possible reason is that the increases in ROS-related enzymes are to reduce the ROS levels in related cells by compensatory mechanisms, and did not result in metabolic or cellular dysfunction.

Although increasing numbers of researchers have focused on the oxidative stress responses arising from xenobiotics, the reports on fish species were very limited and mainly examined the changes in activities of related antioxidant enzymes. For instance, Cazenave et al. (2006) reported that the activities of GR, GSH-PX, and CAT were enhanced in the liver of *Corydoras paleatus* after its exposure to 2 µg L⁻¹ microcystin-RR. More recently, differential expression of the genes encoding these antioxidant proteins has been used to detect biological toxicity and/or to monitor the impact of chemical pollutants (Sheader et al., 2006; Woo et al., 2009). Furthermore, a large number of genes that are related to the oxidative stress response have been identified in zebrafish (Craig et al., 2007; Liu et al., 2008). For this reason, we examined the representative genes of proteins that are used to combat oxidative stress (such as Sod, Cat and Gpx) to determine whether these might serve as molecular endpoints for ATZ exposure in zebrafish. We found that the mRNA levels of Cu/Zn-Sod, Mn-Sod, Cat and Gpx in the liver increased significantly at medium (10 or

100 µg L⁻¹) ATZ exposure concentrations, yet showed no significant amount of induction at the highest treatment concentration (1000 µg L⁻¹). Thus, it could be concluded that these genes' expression could not serve as the effective endpoints for low concentration ATZ exposure in zebrafish. Interestingly, the mRNA induction patterns were not in accordance with the changes in levels of antioxidant enzymes observed in both the liver and ovary. The results suggest that some antioxidant enzymes are controlled by enzyme activity and not by transcriptional controls, at least in the case of SOD and CAT in zebrafish. It is possible that mRNA levels represent a snapshot of the cell activity at any given time, and the protein activity might be regulated at the post-translational level. A similar result was demonstrated in a previous experiment involving oxidative stress in fish. Olsvik et al. (2005) reported that they were not able to link (in a statistically significant manner) hyperoxia to the altered mRNA expression of Cu/Zn-Sod, Cat or Gpx in the liver of smoltifying salmon after long periods (54 or 72 d) of exposure to 130% O₂, even though GSH data suggested that an increase in oxidative stress might have occurred. More recently, Craig et al. (2007) demonstrated that there was no association between increased transcription and enzyme activities related to the antioxidant stress caused by copper exposure in zebrafish.

The liver is reported to be the primary organ for bioaccumulation, and thus, has been extensively studied in regards to the toxic effects of xenobiotics (Jin et al., 2008; Liu et al., 2008). In general, oxidative stress effects were reflected more efficiently and effectively in the liver. For example, Elia et al. (2002) reported that 6 and 9 mg L⁻¹ ATZ exposures induced a significant change in the activities of GPX and SOD and induced a change in the content of MDA and GSH in the liver of bluegill sunfish after 96 h of exposure,

while no response was found in the gill. Thus, the liver was viewed as a primary organ for ATZ metabolism in zebrafish. Furthermore, the mitochondria is the most important subcellular organelle for ROS generation, and the O_2^- content is about 5–10 times higher than that of the cytoplasm. The turbulent ROS generation is related to the mitochondrial respiratory chain and to ATP synthesis. Thus, we attempted to detect the expression of genes related to the mitochondrial inner membrane genes that manage ROS production, the respiratory chain, and ATP synthesis in the liver of zebrafish following ATZ exposure. Correspondingly, we found that Ucp-2 mRNA was induced significantly after ATZ exposure at the $1000 \mu\text{g L}^{-1}$ concentration. Ucp-2 is a member of the inner mitochondrial membrane anion-carrier superfamily. Ucp-2 works, by an incompletely defined mechanism, to reduce the production of reactive oxygen species during mitochondrial electron transport (Giardina et al., 2008). The possible explanation is that an increase in Ucp-2 in the inner mitochondrial membrane will lower mitochondrial O_2^- production and modulate H_2O_2 mitochondrial generation, which is consistent with a mild uncoupling response to oxidative stress (Nègre-Salvayre et al., 1997; Echtaay et al., 2002). Conversely, Bcl-2 expression was inhibited in the liver of zebrafish after ATZ exposure. Bcl-2, which localizes to sites where free radicals are generated, functions as an apparent antioxidant against oxidative stress to prevent apoptosis (Hockenbery et al., 1993). A reduction in Bcl-2 expression, as a result of ATZ exposure, may explain the enhanced oxidative stress in the liver. Previous reports have also shown that oxidative stress actually inhibits Bcl-2 mRNA expression in zebrafish (Liu et al., 2008; Deng et al., 2009). In addition, although ROS production is closely tethered to the mitochondrial respiratory chain, ATZ treatments did not induce any apparent changes in the transcriptional expression of Cx1, Ndi, and ATP6. A similar result was also suggested by Liu et al. (2008), where no significant induction in the above genes was observed in the liver of female zebrafish, which had been exposed to acute perfluorododecanoic acid, even in the presence of heavy oxidative damage.

5. Conclusion

ATZ exposure can induce significant changes in antioxidant enzymes activities (SOD and CAT), and GSH and MDA contents in the tissues of the liver in zebrafish, suggesting that ATZ exposure in female zebrafish promotes an oxidative stress response in the liver. Correspondingly, our study further demonstrated that the mRNA levels of genes, which encode antioxidant proteins (such as Sod, Cat and Gpx) and mitochondrial inner membrane proteins related to ROS production (such as Bcl-2 and Ucp-2), were also significantly altered in the liver of ATZ-treated female zebrafish. To our knowledge, this is the first report to study the transcriptional effects of ATZ stress on zebrafish. Thus, the information presented in this study is helpful to understand the mechanism of ATZ-induced oxidative stress in fish.

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